OurMocks

2027 CYCLE

MOCK · PRACTICE · CRACK

CUET UG

Physics (322)

Ray Optics & Optical Instruments

Official Syllabus Unit 6 · Class 12 NCERT · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA Physics-322, Unit 6)	Skills the questions test
Reflection of light	Spherical mirrors, mirror formula $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$, Cartesian sign convention, magnification $m=-\frac{v}{u}=\frac{h'}{h}$.	Sign discipline, image-position recall, magnification sign reading ($m<0$ real, $m>0$ virtual).
Refraction of light	Refraction through planar boundaries, Snell's laws, apparent depth, normal shift, total internal reflection (TIR) and its critical angle, optical fibres, and totally reflecting prisms.	Applying $n_1\sin\theta_1=n_2\sin\theta_2$, apparent-depth calculations, circle of illuminance, TIR dual criteria.
Refraction at spherical surfaces & lenses	Refraction at spherical surface $\left(\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}\right)$, thin lens formula, lensmaker's formula, power $P=\frac{1}{f}$ (dioptre), combination of thin lenses in contact, medium immersion.	Curvature signs (R_1,R_2), dioptre arithmetic ($P=P_1+P_2$), lens immersion flips, lens cutting rules.
Prisms	Refraction and dispersion through a triangular prism, angle of minimum deviation $\mu=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$, thin-prism relation $\delta\approx(\mu-1)A$, Cauchy's dispersion law, dispersive power.	Index from δ_m, thin-prism shortcuts, rainbow mechanisms, VIBGYOR dispersion ordering.
Optical instruments	Simple microscope (magnifying glass), compound microscope, astronomical refracting telescope, reflecting telescopes (Cassegrain): magnifying powers and tube lengths at normal and near-point adjustments.	Formula recall for M and L, objective-versus-eyepiece roles, resolving power, chromatic aberration elimination.

EXAM PATTERN

CUET UG Pattern (NTA 2026–2027 Baseline): The Physics paper contains 50 compulsory MCQs in 60 minutes (+5 marks for correct, −1 mark for incorrect). Unit 6 (Ray Optics & Optical Instruments) typically yields 4 to 6 questions spanning mirror/lens sign conventions, TIR critical angle conditions, minimum deviation of prisms, and telescope/microscope tube lengths.

Reflection at Spherical Mirrors

A spherical mirror is a small slice cut from a hollow glass sphere, silvered on one side. When the **inner** (concave) surface reflects, the mirror converges parallel incident rays towards a real focal point. When the **outer** (convex) surface reflects, the mirror diverges rays outward, causing them to appear to diverge from a virtual focal point behind the mirror.

Fundamental Geometric Points & Sign Conventions

All analytical problems in geometric optics require strict adherence to the **New Cartesian Sign Convention**:

- **Pole (P):** The geometric centre of the reflecting surface, taken as the coordinate origin $(0, 0)$.
- **Principal Axis:** The straight line passing through the pole P and centre of curvature C.
- **Direction of Light:** Light is assumed to travel from **left to right**. Distances measured in the direction of incident light are **positive**; distances measured **against** incident light are **negative**.
- **Heights:** Heights measured perpendicular to and **above** the principal axis are **positive** $(+y)$; heights **below** are **negative** $(-y)$.
- **Focal Length (f):** For a concave mirror, C and F lie in front (to the left), so $\boldsymbol{f < 0}$ and $\boldsymbol{R < 0}$. For a convex mirror, C and F lie behind (to the right), so $\boldsymbol{f > 0}$ and $\boldsymbol{R > 0}$.
- **Paraxial Ray Relation:** For mirrors of small aperture, the focal length is exactly half the radius of curvature:

$$f = \frac{R}{2}$$

Mirror Formula & Linear Magnification

From geometry of similar right triangles formed by paraxial rays:

MASTER SPHERICAL MIRROR EQUATIONS

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$$

$$m = \frac{h'}{h} = -\frac{v}{u} = \frac{f}{f-u} = \frac{f-v}{f}$$

Physical Interpretation of Signs:

1. Object distance u is virtually always **negative** for real objects $(u < 0)$.
2. Real images are formed by actual intersection of reflected rays in front of the mirror $(v < 0)$; hence magnification $m = -\frac{v}{u} = -\frac{-v}{-u} < 0$ is **negative** (real images are always **inverted**).
3. Virtual images are formed behind the mirror $(v > 0)$; hence magnification $m = -\frac{+v}{-u} > 0$ is **positive** (virtual images are always **erect**).
4. Convex mirrors always produce $|v| < f$ with $v > 0$, yielding **virtual, erect, and diminished** images $(0 < m < 1)$ regardless of object position!

WORKED EXAMPLE — CONCAVE MIRROR IMAGE TRACING

An object of height 2.5 cm is placed at a distance of 30.0 cm in front of a concave mirror of radius of curvature 40.0 cm. Find the position, nature, and height of the image.

Solution: Focal length $f = \frac{R}{2} = \frac{-40.0}{2} = -20.0$ cm. Object distance $u = -30.0$ cm.

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-20} - \frac{1}{-30} = -\frac{1}{20} + \frac{1}{30} = \frac{-3+2}{60} = -\frac{1}{60} \Longrightarrow v = -60.0 \text{ cm}$$

Since $v = -60.0$ cm < 0, the image is **real** and formed 60.0 cm in front of the mirror. Magnification:

$$m = -\frac{v}{u} = -\frac{-60.0}{-30.0} = -2.0$$

Image height $h' = m \times h = (-2.0) \times (2.5 \text{ cm}) = -5.0$ cm (**inverted**, twice object size).

Refraction of Light & Snell's Law

Refraction is the change in the direction of propagation of a light wave as it passes obliquely across the boundary between two transparent media with differing optical densities.

Physical Cause & Invariant Parameters

Light travels at speed $c \approx 3.00 \times 10^8$ m/s in vacuum, but slows down when entering a material medium. The **absolute refractive index** μ (or n) of a medium is defined as:

$$\mu = \frac{c}{v}$$

Where v is the phase speed of light in that medium.

Note:
The Invariance of Frequency: When a light wave crosses an interface between two media, its **frequency ν remains strictly invariant** because frequency is determined solely by the source oscillations ($E = h\nu = \text{const}$). Consequently, because $v = \nu\lambda$, the wavelength **compresses** inside the medium:

$$\lambda' = \frac{\lambda}{\mu}$$

Snell's Law of Refraction

For a light ray incident at angle i in medium 1 (μ_1) and refracting at angle r in medium 2 (μ_2):

SNELL'S LAW OF REFRACTION

$$\mu_1 \sin i = \mu_2 \sin r \Longrightarrow \frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} = {}^1\mu_2 = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

- **Rarer to Denser ($\mu_2 > \mu_1$):** Ray bends **towards** the normal ($r < i$). Speed and wavelength decrease.
- **Denser to Rarer ($\mu_2 < \mu_1$):** Ray bends **away from** the normal ($r > i$). Speed and wavelength increase.
- **Normal Incidence ($i = 0°$):** $r = 0°$. The ray passes undeviated, though its speed and wavelength change.

Real and Apparent Depth & Normal Shift

When an object placed in an optically denser medium (μ) is viewed from an optically rarer medium (air, $\mu_1 = 1$), refracted rays bend away from the normal. When extrapolated backwards, they intersect at a virtual image point I located higher than object O:

$$\text{Apparent depth} \quad d' = \frac{d}{\mu}$$

The **normal shift** Δt by which the object appears raised is:

NORMAL SHIFT FORMULA

$$\Delta t = d - d' = d\left(1 - \frac{1}{\mu}\right)$$

Crucial Property: The normal shift Δt depends **only on slab thickness d and refractive index μ**; it is completely independent of where the object is located beneath the slab! For multiple composite layers of immiscible liquids:

$$\Delta t_{\text{total}} = \sum_i d_i \left(1 - \frac{1}{\mu_i}\right)$$

Lateral Displacement Through a Parallel Glass Slab

A light ray entering a parallel-sided transparent slab refracts at the top interface and again at the bottom interface. Because the two refracting faces are parallel, the emergent ray is **parallel to the incident ray** (net angular deviation $\delta = 0°$), but shifted laterally by distance x:

$$x = \frac{t \sin(i - r)}{\cos r}$$

Where t is the thickness of the slab.

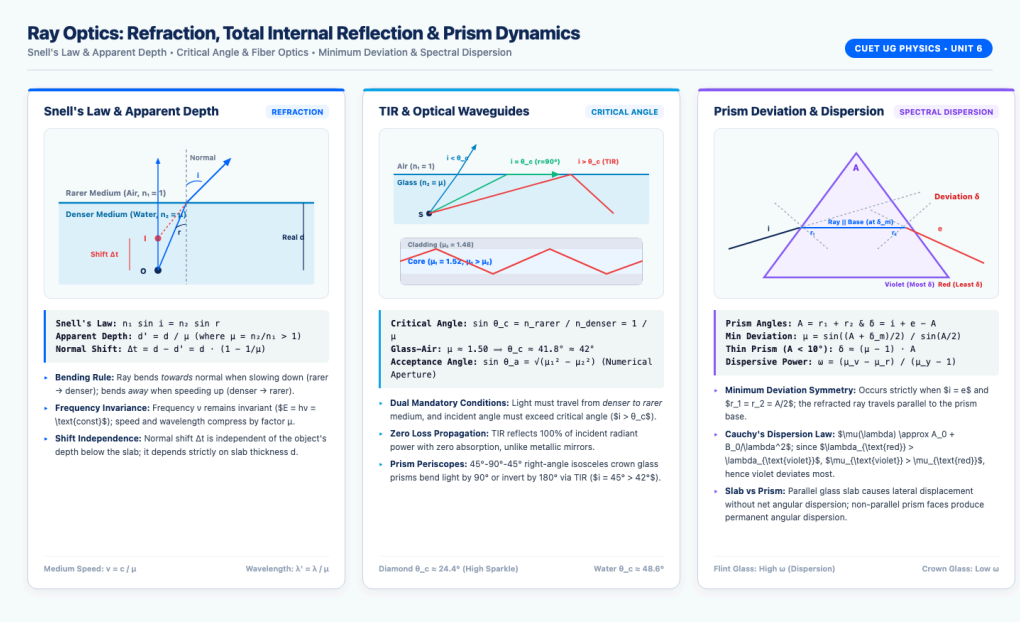


Figure 1: Refraction, Total Internal Reflection & Prism Dynamics: Snell's law geometry, apparent depth shift, critical angle conditions in optical fibres, and minimum deviation symmetry in triangular prisms.

Total Internal Reflection (TIR) & Critical Angle

When a light ray travels from an optically denser medium to an optically rarer medium, the angle of refraction r exceeds the angle of incidence i. As i increases, r eventually reaches 90°. The angle of incidence in the denser medium for which the angle of refraction in the rarer medium equals 90° is called the **critical angle** θ_c.

Dual Mandatory Conditions for TIR

Total internal reflection occurs if and only if **both** conditions are met simultaneously:

1. **Direction Criterion:** Light must attempt to travel from an **optically denser medium to an optically rarer medium** ($\mu_1 > \mu_2$).
2. **Angle Criterion:** The angle of incidence must strictly exceed the critical angle: $\boldsymbol{i > \theta_c}$.

From Snell's law at $i = \theta_c$ and $r = 90°$:

$$\mu_1 \sin\theta_c = \mu_2 \sin 90° \implies \sin\theta_c = \frac{\mu_2}{\mu_1}$$

For a medium-to-air interface ($\mu_2 = 1, \mu_1 = \mu$):

CRITICAL ANGLE FORMULA

$$\sin\theta_c = \frac{1}{\mu} \implies \theta_c = \arcsin\left(\frac{1}{\mu}\right)$$

Standard Benchmark Critical Angles:

- Crown glass ($\mu = 1.50$): $\theta_c = \arcsin\left(\frac{1}{1.50}\right) \approx 41.8° \approx 42°$
- Water ($\mu = 1.33 = \frac{4}{3}$): $\theta_c = \arcsin\left(\frac{3}{4}\right) \approx 48.6°$
- Diamond ($\mu = 2.42$): $\theta_c = \arcsin\left(\frac{1}{2.42}\right) \approx 24.4°$ (extreme sparkle due to multiple internal bounces!)

Circle of Illuminance (Fish-Eye Horizon)

A point source of light S submerged at depth h inside water emits rays in all directions. Only rays incident at the water-air interface with $i \leq \theta_c$ can escape into the air. Rays with $i > \theta_c$ undergo TIR back into the water. Consequently, light emerges only through a circular patch on the surface of radius r:

$$\tan\theta_c = \frac{r}{h} \Longrightarrow r = h\tan\theta_c = h\frac{\sin\theta_c}{\cos\theta_c} = h\frac{\frac{1}{\mu}}{\sqrt{1-\frac{1}{\mu^2}}} = \frac{h}{\sqrt{\mu^2-1}}$$

The area of the luminous circular window is:

$$A = \pi r^2 = \frac{\pi h^2}{\mu^2 - 1}$$

High-Yield Technological Applications of TIR

1. **Optical Fibres:** Consist of a high refractive index dielectric **core** ($\mu_1 \approx 1.52$) surrounded by a lower index **cladding** ($\mu_2 \approx 1.48$). Light entering at angle within the acceptance cone undergoes repeated TIR with virtually **zero transmission loss**.
   - **Numerical Aperture (NA):** Measure of light-gathering capacity:
   $$NA = \sin\theta_a = \sqrt{\mu_1^2 - \mu_2^2}$$
2. **Totally Reflecting Prisms:** Right-angled isosceles crown glass prisms ($45° - 90° - 45°$). Since $\theta_c \approx 42°$, a ray incident normally on one face hits the hypotenuse at $i = 45° > 42°$, undergoing complete TIR:
   - **Deviating by 90°:** Used in periscopes.
   - **Deviating by 180°:** Inverts images without changing size (Porro prisms in binoculars).
   - **Advantage over Silvered Mirrors:** 100% reflection, no ghost reflections, no degradation or tarnishing of reflective coating.
3. **Mirage:** Optical illusion in hot deserts where the air near the heated ground expands and becomes optically rarer than cooler air above (μ decreases towards ground). Rays from tree tops bend away from the normal until $i > \theta_c$, undergoing TIR and entering the eye from below, creating the inverted virtual image of an apparent water pool.

WORKED EXAMPLE — CIRCLE OF ILLUMINANCE

A point source of light is placed at the bottom of a water tank of depth $\sqrt{7}$ m. If the refractive index of water is $\frac{4}{3}$, find the diameter of the circular area on the water surface through which light can emerge.

Solution: Radius of circular patch:

$$r = \frac{h}{\sqrt{\mu^2-1}} = \frac{\sqrt{7}}{\sqrt{\left(\frac{4}{3}\right)^2-1}} = \frac{\sqrt{7}}{\sqrt{\frac{16}{9}-1}} = \frac{\sqrt{7}}{\sqrt{\frac{7}{9}}} = \frac{\sqrt{7}}{\frac{\sqrt{7}}{3}} = 3.0 \text{ m}$$

Therefore, the diameter is $D = 2r = 2 \times 3.0 = 6.0$ m.

Refraction at Spherical Surfaces & Thin Lenses

When light refracts across a curved interface separating two media of indices μ_1 and μ_2, the curved geometry alters wavefront vergence.

Refraction at a Single Spherical Surface

For paraxial rays incident from medium 1 (μ_1) onto a spherical surface of radius R bounding medium 2 (μ_2):

SINGLE SPHERICAL SURFACE REFRACTION EQUATION

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Sign Discipline: Apply Cartesian conventions to u, v, and R. If the surface is convex towards the rarer medium, $R > 0$; if concave towards the rarer medium, $R < 0$.

Lensmaker's Formula & Thin Lens Equation

Applying the single-surface equation across two consecutive curved surfaces of radii R_1 and R_2 bounding a lens of material index $\mu_2 = \mu_g$ surrounded by medium $\mu_1 = \mu_m$:

LENSMAKER'S FORMULA

$$\frac{1}{f} = \left(\frac{\mu_g}{\mu_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For a lens in air ($\mu_m = 1, \mu_g = \mu$):

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Combining with the object-image relation yields the **Gaussian Thin Lens Equation**:

GAUSSIAN THIN LENS FORMULA & MAGNIFICATION

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$m = \frac{h'}{h} = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$$

- **Convex (Converging) Lens:** Thicker at centre; $f > 0$; forms real inverted images for $u > f$ and virtual erect magnified images for $u < f$.
- **Concave (Diverging) Lens:** Thinner at centre; $f < 0$; always forms virtual, erect, and diminished images between optical centre and focus ($0 < m < 1$).

Power of a Lens & Combination in Contact

The **power** P of a lens measures its ability to converge or diverge incident light:

$$P = \frac{1}{f \text{ (in metres)}} = \frac{100}{f \text{ (in cm)}}$$

SI Unit of power is the **Dioptre (D)**, where $1\ \text{D} = 1\ \text{m}^{-1}$.

- Converging lens: $f > 0 \Longrightarrow P > 0$ (positive power).
- Diverging lens: $f < 0 \Longrightarrow P < 0$ (negative power).

When two or more thin lenses of focal lengths $f_1, f_2, \ldots$ and powers $P_1, P_2, \ldots$ are placed in **coaxial contact**:

LENSES IN CONTACT

$$P_{\text{eq}} = P_1 + P_2 + P_3 + \ldots$$

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + \ldots$$

$$m_{\text{eq}} = m_1 \times m_2 \times m_3 \times \ldots$$

Liquid Immersion & Lens Inversion

When a glass lens (μ_g) of focal length f_{air} in air is immersed in a liquid of refractive index μ_l:

$$\frac{\frac{1}{f_{\text{liq}}}}{\frac{1}{f_{\text{air}}}} = \frac{\left(\frac{\mu_g}{\mu_l} - 1\right)}{\mu_g - 1} \Longrightarrow f_{\text{liq}} = f_{\text{air}} \frac{(\mu_g - 1)}{\frac{\mu_g}{\mu_l} - 1}$$

Three Dramatic Immersion Cases Tested by CUET:

1. **Immersion in Water ($\mu_l < \mu_g$):** For glass ($\mu_g = 1.5$) in water ($\mu_l = \frac{4}{3}$):

$$f_{\text{water}} = f_{\text{air}} \frac{1.5-1}{\frac{1.5}{\frac{4}{3}}-1} = \frac{0.5}{\frac{9}{8}-1} f_{\text{air}} = \frac{0.5}{\frac{1}{8}} f_{\text{air}} = 4f_{\text{air}}$$

The focal length increases by a factor of 4; power drops to $\frac{1}{4}$th of its air value.

2. **Immersion in Matching Medium ($\mu_l = \mu_g$):** $\frac{1}{f_{\text{liq}}} = 0 \Longrightarrow f_{\text{liq}} \to \infty$. The lens behaves as a plain glass sheet and becomes completely **invisible**!
3. **Immersion in Denser Liquid ($\mu_l > \mu_g$):** For glass ($\mu_g = 1.5$) in carbon disulphide ($\mu_l = 1.63$): $\left(\frac{\mu_g}{\mu_l} - 1\right) < 0$. The sign of focal length **flips**! A convex lens becomes **diverging**, and a concave lens becomes **converging**!

Lens Cutting Dynamics

- **Cut Along Principal Axis (Longitudinal):** Radii R_1, R_2 and $(\mu - 1)$ are unaltered. Focal length of each half remains $f' = f$, power remains $P' = P$, but aperture area is halved, so image **intensity is halved** ($I' = \frac{I}{2}$).
- **Cut Perpendicular to Principal Axis (Transverse):** An equiconvex lens ($R_1 = R, R_2 = -R$) has $\frac{1}{f} = (\mu - 1)\left(\frac{2}{R}\right)$. Each resulting plano-convex half has surfaces R and ∞, so $\frac{1}{f'} = (\mu - 1)\left(\frac{1}{R}\right) = \frac{1}{2f}$. The focal length **doubles** ($f' = 2f$) and power is **halved** ($P' = \frac{P}{2}$).

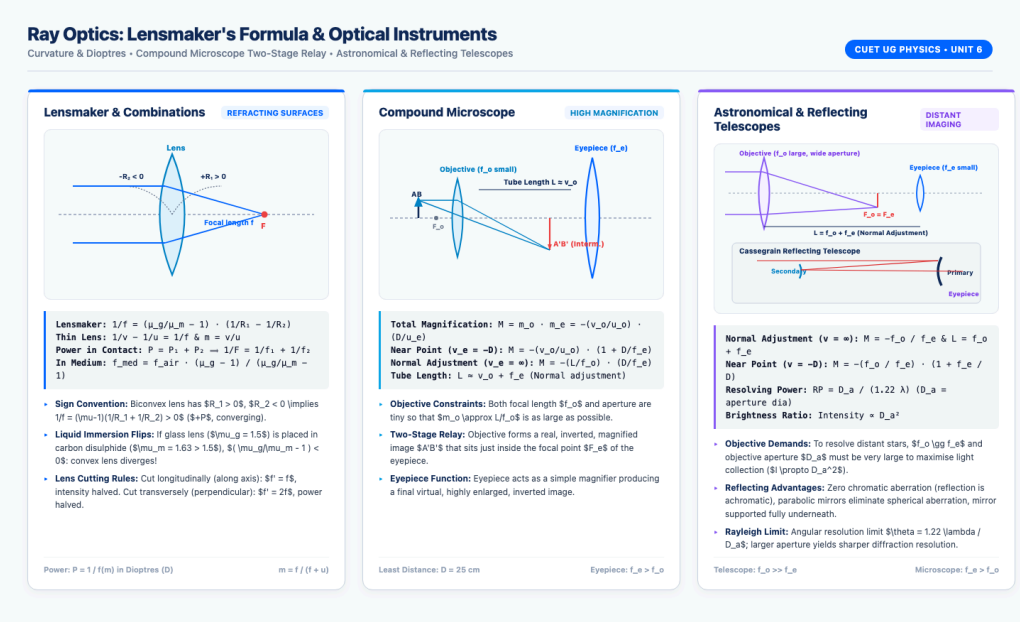


Figure 2: Lensmaker's Formula & Optical Instruments: Biconvex lens curvature and focal convergence, compound microscope two-stage magnification relay, and astronomical vs Cassegrain reflecting telescopes.

Prisms, Minimum Deviation & Dispersion

A prism is a homogeneous, transparent optical solid bounded by two non-parallel plane refracting surfaces inclined at the **angle of the prism** A (or refracting angle).

Geometry of Prism Deviation

A monochromatic ray incident at angle i on the first face refracts at angle r_1, travels through the prism, strikes the second face at angle r_2, and emerges at angle e. From cyclic quadrilateral and exterior triangle geometry:

MASTER PRISM RELATIONS

$$A = r_1 + r_2$$
$$\delta = i + e - A$$

Angle of Minimum Deviation (δ_m)

As the angle of incidence i is gradually increased, the deviation δ decreases, passes through a distinct minimum δ_m, and then increases. At the condition of **minimum deviation**:

1. The ray passes symmetrically through the prism: $\boldsymbol{i = e}$.
2. The internal refraction angles are equal: $\boldsymbol{r_1 = r_2 = r = \frac{A}{2}}$.
3. For an equilateral or isosceles prism, the refracted ray inside travels strictly **parallel to the base**.

Substituting into the prism equations:

$$A = r + r = 2r \Longrightarrow r = \frac{A}{2}$$

$$\delta_m = i + i - A = 2i - A \Longrightarrow i = \frac{A + \delta_m}{2}$$

Applying Snell's law at the first refracting face gives the famous **Prism Formula**:

PRISM FORMULA FOR REFRACTIVE INDEX

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Thin Prism Approximation & Dispersion

For a thin prism with small refracting angle ($A < 10°$), using $\sin\theta \approx \theta$:

$$\delta \approx (\mu - 1)A$$

Dispersion of White Light: Because refractive index varies with wavelength according to **Cauchy's Empirical Formula**:

$$\mu(\lambda) \approx A_0 + \frac{B_0}{\lambda^2}$$

Since $\lambda_{\text{violet}} < \lambda_{\text{red}}$, the refractive index is highest for violet and lowest for red:

$$\mu_{\text{violet}} > \mu_{\text{yellow}} > \mu_{\text{red}}$$

Consequently, violet deviates the most and red deviates the least ($\delta_v > \delta_r$).

- **Angular Dispersion (θ):** The angular separation between extreme rays (violet and red):
$$\theta = \delta_v - \delta_r = (\mu_v - \mu_r)A$$
- **Mean Deviation (δ_y):** Deviation experienced by mean yellow light:
$$\delta_y = (\mu_y - 1)A$$
- **Dispersive Power (ω):** Ratio of angular dispersion to mean deviation:

  DISPERSIVE POWER

  $$\omega = \frac{\theta}{\delta_y} = \frac{\mu_v - \mu_r}{\mu_y - 1}$$

  Key Exam Fact: Dispersive power ω depends **strictly on the material** of the prism, and is completely independent of the prism angle A!

Optical Instruments: Microscopes & Telescopes

Optical instruments are multi-element systems designed to increase the **visual angle** subtended by tiny or distant objects at the human eye.

The Human Eye & Visual Angle

The perceived size of an object is determined solely by the angle subtended by it at the eye (visual angle). The closest distance at which a normal eye can focus clearly without strain is the **least distance of distinct vision**, $D = 25$ cm (near point). At infinity, the eye muscles are completely relaxed (normal adjustment).

Simple Microscope (Magnifying Glass)

A single converging lens of short focal length f. The object is placed within the focal length ($u < f$), forming an erect, virtual, and magnified image on the same side.

- **Image at Near Point ($v = -D = -25$ cm):**

$$M = 1 + \frac{D}{f}$$

- **Image at Infinity (Normal Adjustment / Relaxed Eye):**

$$M = \frac{D}{f}$$

Compound Microscope

Employs two coaxial convex lenses to achieve magnifications beyond the aberrations of a single lens:

1. **Objective Lens:** Very small focal length (f_o) and tiny aperture. The object is placed just beyond F_o ($u_o > f_o$), forming a **real, inverted, and magnified intermediate image** $A'B'$.
2. **Eyepiece (Ocular):** Moderate focal length ($f_e > f_o$) and larger aperture. Positioned such that $A'B'$ falls within its focal length, acting as a simple magnifier to form a **final virtual, inverted (relative to original object), and highly magnified image** $A''B''$.

COMPOUND MICROSCOPE MAGNIFYING POWER

$$M = m_o \times m_e = -\left(\frac{v_o}{u_o}\right) \times \left(\frac{D}{u_e}\right)$$

$$\text{At Near Point } (\, v_e = -D \,): \quad M = -\left(\frac{v_o}{u_o}\right)\left(1 + \frac{D}{f_e}\right) \approx -\left(\frac{L}{f_o}\right)\left(1 + \frac{D}{f_e}\right)$$

$$\text{At Normal Adjustment } (\, v_e = \infty \,): \quad M = -\left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right)$$

$$\text{Tube Length } (\, v_e = \infty \,): \quad L \approx v_o + f_e$$

Where L is the distance between the second focal point of the objective and the first focal point of the eyepiece.

Astronomical Refracting Telescope

Designed to view distant astronomical objects ($u \approx \infty$):

1. **Objective Lens:** Large focal length (f_o) and very large aperture to maximize light collection and diffraction resolution. Forms a **real, inverted, diminished intermediate image** at its focal plane F_o.
2. **Eyepiece:** Short focal length ($f_e < f_o$) and small aperture. Magnifies the intermediate image.

ASTRONOMICAL TELESCOPE FORMULAS

$$\text{At Normal Adjustment } (\, v_e = \infty \,): \quad M = -\frac{f_o}{f_e} \quad \text{and} \quad L = f_o + f_e$$

$$\text{At Near Point } (\, v_e = -D \,): \quad M = -\frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right) \quad \text{and} \quad L = f_o + u_e$$

Reflecting Telescopes (Cassegrain Design)

Modern astronomical observatories exclusively utilize reflecting telescopes with large parabolic concave primary mirrors and convex secondary mirrors. **Four Decisive Advantages over Refracting Telescopes:**

1. **Zero Chromatic Aberration:** Reflection obeys the law of reflection ($i = r$) identically for all wavelengths, eliminating chromatic aberration entirely.
2. **Elimination of Spherical Aberration:** Parabolic mirror geometry focuses all parallel rays to an identical mathematical point.

3. **Mechanical Rigidity:** Heavy mirrors can be supported mechanically across their entire rear back, whereas massive refracting lenses can only be supported along their rims, causing gravitational sagging.
4. **Giant Light-Gathering Power:** High-diameter apertures ($D_a > 10$ m) gather vast amounts of light ($I \propto D_a^2$) and provide razor-sharp angular resolution:

$$\theta = \frac{1.22\lambda}{D_a}$$

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Master Mindmap: Ray Optics & Optical Instruments

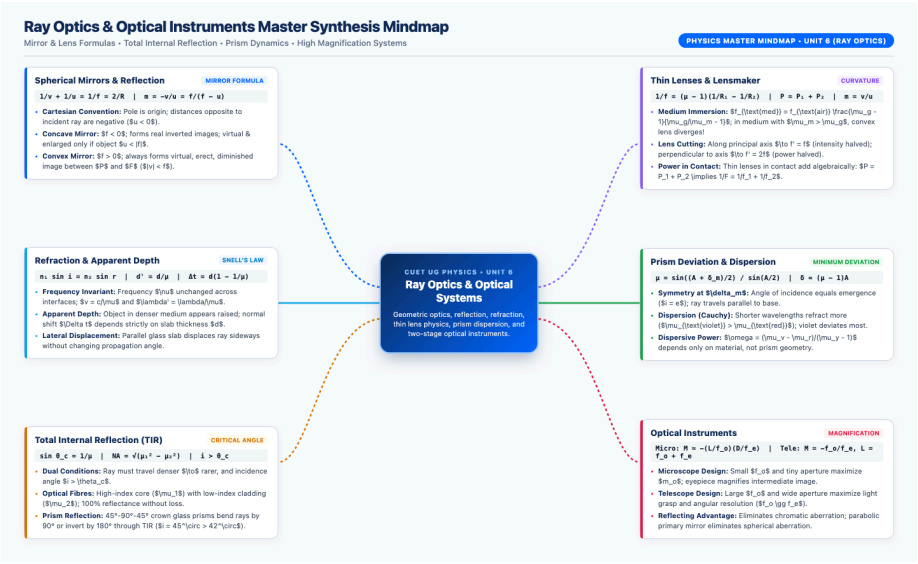


Figure 3: Master Synthesis Mindmap: Structural framework of Unit 6 Ray Optics and Optical Instruments.

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High-Yield Cheatsheet & Exam Traps

MASTER RAY OPTICS & OPTICAL INSTRUMENTS REVISION CHEATSHEET

- **Mirror Equation:** $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}=\frac{2}{R}$ | Magnification: $m=-\frac{v}{u}=\frac{f}{f-u}$.
- **Snell's Law:** $\mu_1 \sin i=\mu_2 \sin r$ | Apparent depth: $d'=\frac{d}{\mu}$ | Normal shift: $\Delta t=d\left(1-\frac{1}{\mu}\right)$.
- **Critical Angle:** $\sin\theta_c=\frac{\mu_2}{\mu_1}=\frac{1}{\mu}$ | Circle of illuminance: $r=\frac{h}{\sqrt{\mu^2-1}}$ | Area: $A=\frac{\pi h^2}{\mu^2-1}$.
- **Single Surface Refraction:** $\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$.
- **Lensmaker's Formula:** $\frac{1}{f}=\left(\frac{\mu_g}{\mu_m}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$ | Lens in air: $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$.
- **Thin Lens Equation:** $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ | Power: $P=\frac{1}{f}$ (m) $=\frac{100}{f}$ (cm) | Contact: $P=P_1+P_2$.
- **Liquid Immersion:** Glass ($\mu=1.5$) in water ($\mu=\frac{4}{3}$): $f_w=4f_a$. In matching liquid ($\mu_l=\mu_g$): $f\to\infty$ (invisible). In denser liquid ($\mu_l>\mu_g$): lens character reverses!
- **Lens Cutting:** Longitudinally: $f'=f, P'=P, I'=\frac{I}{2}$. Transversely: $f'=2f, P'=\frac{P}{2}$.
- **Prism Refraction:** $A=r_1+r_2$ | $\delta=i+e-A$ | Min deviation: $\mu=\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$.
- **Thin Prism:** $\delta\approx(\mu-1)A$ | Dispersion: $\theta=(\mu_v-\mu_r)A$ | Dispersive power: $\omega=\frac{\mu_v-\mu_r}{\mu_y-1}$.
- **Compound Microscope:** $M_\infty=-\left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right)$ | $L=v_o+f_e$ | Near point: $M_D=-\left(\frac{v_o}{u_o}\right)\left(1+\frac{D}{f_e}\right)$.
- **Astronomical Telescope:** $M_\infty=-\frac{f_o}{f_e}$ | $L=f_o+f_e$ | Near point: $M_D=-\left(\frac{f_o}{f_e}\right)\left(1+\frac{f_e}{D}\right)$.

THE SEVEN FATAL CUET TRAPS IN RAY OPTICS

1. **The Mirror vs Lens Sign Trap:** Mirror formula is $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$ with $m=-\frac{v}{u}$; Lens formula is $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ with $m=+\frac{v}{u}$. Mixing signs is the single most common source of -1 marks!
2. **The Normal Shift Independence Trap:** Normal shift $\Delta t=d\left(1-\frac{1}{\mu}\right)$ depends **strictly on slab thickness d and index μ**. It does **not** depend on the distance of the object from the slab!
3. **TIR Direction Criterion Trap:** Total internal reflection can **never** occur when light enters from air to glass or water to glass. The ray **must** travel from denser to rarer medium with $i>\theta_c$.
4. **The Liquid Immersion Inversion Trap:** Immersing a convex glass lens ($\mu=1.5$) into carbon disulphide ($\mu=1.63$) does not merely change focal length—it turns the lens into a **diverging** lens!
5. **Lens Cutting Trap:** Cutting an equiconvex lens along its principal axis preserves f ($f'=f$); cutting perpendicular to its principal axis doubles f ($f'=2f$) and halves power!
6. **Prism Minimum Deviation Base-Parallel Trap:** The ray inside a prism travels parallel to the base **only** when the prism is equilateral or isosceles **and** operating at minimum deviation ($i=e$).
7. **Telescope vs Microscope Objective Trap:** A telescope requires a **large** f_o and large aperture for high M and resolution; a microscope requires a **very small** f_o to maximize linear magnification $m_o\approx\frac{L}{f_o}$!

CUET UG Exam-Style Practice Questions (15 MCQs)

Q1. An object is placed at a distance of 30.0 cm in front of a concave mirror of focal length 20.0 cm. The image formed is:

(A) Virtual, erect, at 60.0 cm behind mirror
(B) Virtual, erect, at 12.0 cm behind mirror
(C) Real, inverted, at 12.0 cm in front of mirror
(D) Real, inverted, at 60.0 cm in front of mirror

Sol. Q1. $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-20} - \frac{1}{-30} = -\frac{1}{60} \Rightarrow v = -60.0$ cm. $m = -\frac{v}{u} = -\frac{-60}{-30} = -2.0$. Real, inverted, magnified image at 60.0 cm in front.
Answer: (D)

Q2. A convex mirror of focal length f forms an image of a real object placed at distance u. Which statement is always true?

(A) Image is real and formed beyond $2f$
(B) Image is virtual, erect, and $|v| < f$
(C) Image is virtual and magnified if $u < f$
(D) Image is inverted and at C when $u = 2f$

Sol. Q2. For a convex mirror ($f > 0$), $\frac{1}{v} = \frac{1}{f} + \frac{1}{|u|} > \frac{1}{f} \Rightarrow 0 < v < f$ for all real object distances $u < 0$. The image is always virtual, erect, and diminished.
Answer: (B)

Q3. Monochromatic light of wavelength 600 nm in vacuum enters a glass plate of refractive index 1.50. Its frequency and wavelength inside the glass are:

(A) 5.0×10^{14} Hz and 600 nm
(B) 3.33×10^{14} Hz and 400 nm
(C) 5.0×10^{14} Hz and 400 nm
(D) 7.5×10^{14} Hz and 900 nm

Sol. Q3. Frequency is invariant: $\nu = \frac{c}{\lambda} = \frac{3.0\times10^{8}}{600\times10^{-9}} = 5.0 \times 10^{14}$ Hz. Wavelength in glass: $\lambda' = \frac{\lambda}{\mu} = \frac{600}{1.50} = 400$ nm.
Answer: (C)

Q4. A glass plate ($\mu = 1.50$) of thickness 6.0 cm is placed over a stamp. By what distance does the stamp appear raised when viewed normally?

(A) 4.0 cm
(B) 1.5 cm
(C) 2.0 cm
(D) 3.0 cm

Sol. Q4. Normal shift $\Delta t = d\left(1 - \frac{1}{\mu}\right) = 6.0\left(1 - \frac{1}{1.50}\right) = 6.0\left(\frac{1}{3}\right) = 2.0$ cm.
Answer: (C)

Q5. A point source of light is at the bottom of a water tank ($\mu = \frac{4}{3}$) of depth $\sqrt{7}$ m. The area of the surface through which light emerges is:

(A) 9π m^2
(B) 7π m^2
(C) 16π m^2
(D) 4π m^2

Sol. Q5. Radius of circle: $r = \frac{h}{\sqrt{\mu^2 - 1}} = \frac{\sqrt{7}}{\sqrt{\frac{16}{9} - 1}} = 3.0$ m. Area $A = \pi r^2 = \pi(3)^2 = 9\pi$ m^2.
Answer: (A)

Q6. In an optical fibre, light is guided along the core by total internal reflection. If the core has refractive index $\mu_1 = 1.50$ and cladding has $\mu_2 = 1.414 = \sqrt{2}$, the critical angle at the core-cladding interface is:

(A) $30°$ **(B)** $45°$
(C) $\arcsin\left(\frac{\sqrt{2}}{1.50}\right) \approx 70.5°$ **(D)** $60°$

Sol. Q6. Condition for TIR at core-cladding boundary: $\sin\theta_c = \frac{\mu_2}{\mu_1} = \frac{\sqrt{2}}{1.50} = \frac{1.414}{1.50} \approx 0.9428 \Longrightarrow \theta_c = \arcsin\left(\frac{\sqrt{2}}{1.50}\right) \approx 70.5°$.

Answer: (C)

Q7. A biconvex lens of glass ($\mu = 1.50$) has radii of curvature $R_1 = 20.0$ cm and $R_2 = -30.0$ cm. Its focal length in air is:

(A) $+12.0$ cm **(B)** $+15.0$ cm
(C) $+50.0$ cm **(D)** $+24.0$ cm

Sol. Q7. Lensmaker's formula: $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.5 - 1)\left(\frac{1}{20} - \left(-\frac{1}{30}\right)\right) = 0.5 \times \frac{3+2}{60} = 0.5 \times \frac{5}{60} = \frac{5}{120} = \frac{1}{24} \Longrightarrow f = +24.0$ cm.

Answer: (D)

Q8. A convex lens of focal length 20 cm in air ($\mu_g = 1.50$) is completely submerged in water ($\mu_w = \frac{4}{3}$). Its new focal length in water is:

(A) 20 cm **(B)** 40 cm
(C) 80 cm **(D)** 5 cm

Sol. Q8. $f_w = f_a \times \frac{(\mu_g - 1)}{\frac{\mu_g}{\mu_w} - 1} = 20 \times \frac{1.5-1}{\frac{1.5}{\frac{4}{3}} - 1} = 20 \times \frac{0.5}{\frac{9}{8} - 1} = 20 \times \frac{0.5}{\frac{1}{8}} = 20 \times 4 = 80$ cm.

Answer: (C)

Q9. Two thin lenses of powers $+5.0$ D and -3.0 D are placed in coaxial contact. The focal length of the combination is:

(A) $+50.0$ cm **(B)** -50.0 cm
(C) $+20.0$ cm **(D)** $+25.0$ cm

Sol. Q9. Net power $P = P_1 + P_2 = +5.0 - 3.0 = +2.0$ D. Equivalent focal length $F = \frac{1}{P} = \frac{1}{2.0} = 0.50$ m $= +50.0$ cm.

Answer: (A)

Q10. An equiconvex lens of focal length f and power P is cut into two identical plano-convex halves along a plane perpendicular to its principal axis. The focal length and power of each half are:

(A) $f' = f, P' = P$ **(B)** $f' = 2f, P' = \frac{P}{2}$
(C) $f' = \frac{f}{2}, P' = 2P$ **(D)** $f' = 4f, P' = \frac{P}{4}$

Sol. Q10. For equiconvex lens: $\frac{1}{f} = (\mu - 1)\left(\frac{2}{R}\right)$. For each plano-convex half: $\frac{1}{f'} = (\mu - 1)\left(\frac{1}{R}\right) = \frac{1}{2f}$. Hence focal length doubles: $f' = 2f$, and power is halved: $P' = \frac{P}{2}$.

Answer: (B)

Q11. An equilateral triangular glass prism ($A = 60°$) produces an angle of minimum deviation $\delta_m = 30°$. The refractive index of the prism material is:

(A) 1.50 **(B)** 1.33
(C) $\sqrt{3} \approx 1.732$ **(D)** $\sqrt{2} \approx 1.414$

Sol. Q11. Prism formula: $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{60°+30°}{2}\right)}{\sin\left(60\frac{°}{2}\right)} = \frac{\sin 45°}{\sin 30°} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} \approx 1.414$.

Answer: (D)

Q12. A thin prism of refracting angle $A = 6.0°$ has refractive indices $\mu_v = 1.53$ and $\mu_r = 1.51$ for violet and red light. The angular dispersion produced is:

(A) 0.12° **(B)** 0.02°
(C) 3.12° **(D)** 0.06°

Sol. Q12. Angular dispersion $\theta = \delta_v - \delta_r = (\mu_v - \mu_r)A = (1.53 - 1.51) \times 6.0° = 0.02 \times 6.0° = 0.12°$.

Answer: (A)

Q13. In an astronomical telescope in normal adjustment, the focal lengths of objective and eyepiece are 100 cm and 5.0 cm respectively. The magnifying power and tube length are:

(A) $M = -20$ and $L = 95$ cm **(B)** $M = -20$ and $L = 105$ cm
(C) $M = -500$ and $L = 105$ cm **(D)** $M = -25$ and $L = 100$ cm

Sol. Q13. At normal adjustment: Magnifying power $M = -\frac{f_o}{f_e} = -\frac{100}{5.0} = -20$. Tube length $L = f_o + f_e = 100 + 5.0 = 105$ cm.

Answer: (B)

Q14. A compound microscope has an objective of focal length 1.0 cm and eyepiece of focal length 5.0 cm. An object is placed at 1.1 cm from the objective. If the final image is formed at the least distance of distinct vision ($D = 25$ cm), the magnifying power is:

(A) -60 **(B)** -50
(C) -30 **(D)** -120

Sol. Q14. For objective: $\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o} = \frac{1}{1.0} + \frac{1}{-1.1} = 1 - \frac{10}{11} = \frac{1}{11} \Longrightarrow v_o = +11$ cm. Linear magnification $m_o = \frac{v_o}{u_o} = \frac{11}{-1.1} = -10$. Eyepiece magnification: $m_e = 1 + \frac{D}{f_e} = 1 + \frac{25}{5.0} = 1 + 5 = 6$. Total magnifying power $M = m_o \times m_e = (-10) \times 6 = -60$.

Answer: (A)

Q15. Which of the following is the primary advantage of a Cassegrain reflecting telescope over a refracting telescope of identical aperture?

(A) Reflecting telescopes have higher chromatic aberration but lower cost

(B) Reflecting telescopes completely eliminate chromatic aberration and parabolic primary mirrors remove spherical aberration

(C) Reflecting telescopes produce upright images without needing an eyepiece

(D) Reflecting telescopes do not require mechanical tube structures

Sol. Q15. Reflection obeys the law of reflection identically for all wavelengths, completely eliminating chromatic aberration. Furthermore, parabolic mirrors eliminate spherical aberration, and large mirrors can be supported over their entire back.

Answer: (B)